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Tools for Improving the Effectiveness of Academic Partnerships in Informing Conservation Practices

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ABSTRACT: To identify effective strategies for managing and enhancing partnerships between conservation organizations (CO) and academic researchers, we interviewed 11 Canadian environmental nongovernmental and governmental organizations that manage conservation lands. Conservation organizations were asked to describe their strategies for setting research priorities, finding research partners, providing incentives, specifying and obtaining deliverables, applying results, and measuring the success of partnerships with academic researchers. Several effective strategies were identified for enhancing the success of academic partnerships. Many COs develop lists of internal research priorities to communicate to the research community beyond their existing networks. Funding is widely viewed as the most effective incentive; however, most COs are limited in the amount of direct research funding they can provide. Instead, they rely on alternative incentives, including providing access to land and data, accommodations at research stations, equipment, and expertise. Peer-reviewed articles are often the most desirable deliverables; however, alternate deliverables are usually welcomed by COs. These include reports, data sets, literature reviews, and workshops or seminars where researchers share knowledge directly with practitioners. Establishing written contracts for deliverables and following up by phone or email helps to ensure that deliverables are received. Participation in research by CO practitioners serving on student committees or as coauthors helps to keep research relevant to COs' needs. COs can develop systems to track and apply research conducted in partnership with academics, including developing records for completed projects, and disseminating research results beyond the project team.

Index terms: effectiveness, evidence, funding, priorities, research

INTRODUCTION

Effective conservation work is informed by the best available scientific information, including knowledge of the condition of biodiversity features, the threats facing these features, and the effectiveness of conservation actions to restore features or reduce threats. Often, the desired empirical data are unavailable for the specific systems and situations for which conservation actions are required. Conservation organizations (COs) focus entirely or partly on species, biodiversity, or habitat conservation and include both governmental and environmental nongovernmental organizations (ENGOS). Though some COs conduct research internally to fill key knowledge gaps, many rely on research conducted by external partners. These partners can include governmental agencies, other ENGOS, or contractors, but frequently involve academic researchers.

Conservation organizations are increasingly expected to base their practices on the best available science. Interest in evidence-based conservation continues to rise, as demonstration of the efficiency and effectiveness of conservation actions helps to justify allocation of limited resources. Conservation practitioners have been criticized for favoring experience over evidence in conservation decision-making (Sutherland et al. 2004), owing to numer-

ous factors including limited funding and restricted access to the primary literature (Prendergast et al. 1999). Calls for evidence-based conservation are increasingly being acted upon (Pullin and Knight 2009) and numerous recommendations exist for modifying the practices and reward structures of conservation scientists and research institutions (Knight et al. 2008; Arlettaz et al. 2010; Cook et al. 2013). The role of practitioners in facilitating a high-impact research agenda has been less frequently addressed; yet practitioners have numerous opportunities to contribute knowledge and incentives to support and enhance conservation science.

Academic researchers have considerable knowledge of interest to COs, and partnerships can be beneficial for both parties. Expertise in particular fields, along with a strong command of the scientific process, can make academic researchers ideal partners for answering specific research questions. Additionally, academic researchers are typically skilled at leveraging limited funding to produce research deliverables by combining matching funding sources, spending funds judiciously, and employing relatively inexpensive students as participants in, or even leaders of, research projects. The engagement of students further aids the conservation community by training the next generation of practitioners and researchers.

Though many advantages exist to conducting conservation research in academic settings, there are also challenges in ensuring that outcomes are relevant to applied conservation work. Disconnects often occur between what is valued in academic circles, where basic research predominates, and the type of specific, applied research most often sought by conservation practitioners (Knight et al. 2008; Cooke 2011). For example, academic reward structures do not always favor applied work, particularly when the work involves duties beyond publication and obtaining grants, or does not culminate in papers published in high-impact journals. The eagerness of students, coupled with the low cost to support them, make them highly attractive contributors to research projects; however, their lack of experience can lead to extensive training requirements, and mistakes or misguided efforts are commonplace. Short time horizons (one year for honors students, two to five years for graduate students) can also be limiting, given the lengthy time scales associated with many ecological processes. Additionally, academic research can proceed too slowly for practitioners who often need results within months rather than the years it can take to conduct and publish research. Mission drift can also be a concern for COs, given that ecological research findings often vary from initial objectives. In such cases, projects originally meant to address specific applied questions may be refocused on theoretical aspects, or on applied elements that are of less immediate value to conservation practitioners.

Despite these limitations, the basic and applied conservation research conducted through academic institutions is of immense value to conservation practitioners. Many COs actively seek partnerships with academic researchers and help to fund this work directly or through in-kind contributions. Numerous strategies exist to maximize the effectiveness of academic partnerships and to ensure that limited conservation resources are used wisely and focused on high-priority applied conservation issues.

The objective of this paper is to provide both practitioners and researchers insight on opportunities for enhancing partnerships

between COs and academic researchers. It provides environmental practitioners a series of options for enhancing the effectiveness of such partnerships in supporting on-the-ground conservation work, based on a series of discussions with Canadian COs. Information from these discussions is summarized below and examples of effective strategies are provided.

METHODS

Staff members of the Nature Conservancy of Canada (NCC) conducted a series of interviews between June and August 2013 with representatives from 11 Canadian land-management and conservation organizations, including ENGOs, governmental organizations, and internal (to NCC) conservation practitioners. Due to limited resources, only Canadian COs were interviewed, and a subset of these was identified manually by the authors to represent a variety of geographies and CO types. Questions were distributed to participants prior to each interview (Appendix), which lasted 1–2 hours.

RESULTS

Setting Research Priorities

Although some of the interviewed COs had specific priority-setting methods or programs, the majority did not formally set research priorities, at least at an organizational scale. Most COs, instead, identified research priorities opportunistically. Often, priorities were identified by individual program managers or working groups and communicated directly to researchers without being formally recorded or vetted at an organizational level. Research priorities were also frequently identified through conservation planning processes, and thereby documented for particular projects at a variety of scales. Such documents were prepared internally or through working groups with external partners and stakeholders. Whether priorities were set informally or in formal planning documents, there was often no central effort to compile this information and identify organization-wide, or broader, conservation community priorities. Organizations that

had developed more formal priority-setting methods found great value in doing so.

Benefits to formalizing the process of setting research priorities include vetting individual perspectives through consensus or group decision-making, identifying topics that are high priorities across the organization, and ranking research questions according to importance and urgency. This process can help COs to formalize their research needs and target limited research resources on timely questions that are of widespread importance.

Examples of successful priority-setting strategies:

- The Parks and Protected Areas Policy Section of the Ontario Ministry of Natural Resources and Forestry recently canvassed 125 ecologists, superintendents, and other staff to identify major research needs in their protected areas and compiled a summary of staff-identified research priorities. Additionally, monthly conference calls with partners through the Ontario Parks Ecologist Network allowed for discussion of research needs.
- The Land Management Branch of Alberta Tourism, Parks and Recreation developed a formal science plan in 2010 and developed research themes through a provincial workshop.
- The Washington Wildlife Habitat Connectivity Working Group (waconnected.org) is a transboundary organization that identifies conservation research priorities at a regional scale.
- Ducks Unlimited Canada annually convenes a team of conservation managers and science leaders who have solicited research needs from their staff. Research needs are then identified for local and regional scales.

Communicating Research Priorities and Needs

COs varied considerably in the degree to which they communicated their research priorities. Research priorities were often presented directly to individual researchers with whom a working relationship already existed. Such researchers were usually

identified as suitable partners through existing formal or informal networks. Organizations that had developed lists of research priorities (either in individual plans or organization-wide lists) often published these on their website or in newsletters. Few COs actively marketed or distributed lists of priorities to the research community.

Efforts to communicate research priorities can provide numerous benefits. Advertising priorities can help to broaden the pool of interested researchers, leading to increased opportunities for matching topics with researchers and allowing greater selectivity in choosing partners. Lists of priority topics can inspire researchers seeking new opportunities or ideas and can help those already working on these topics to share results that were previously unknown to the organization. Ultimately, the communication of research priorities may help to identify and develop partnerships and synergies with other COs, researchers, and even funding agencies with similar research objectives.

Additional opportunities exist to actively communicate (i.e., market) documented research priorities. Social media and research-based listservs (e.g., for professional research associations) can be used to distribute lists of research priorities. Researchers in the organization's network can also be asked to circulate the list of priorities more broadly among their peers.

Examples of approaches for communicating research priorities:

- The BC protected areas research forum (<http://www.unbc.ca/bc-protected-area-research-forum>) provides a venue where practitioners such as the Parks and Protected Areas Program of British Columbia's Ministry of Environment formally present their research priorities to participants that include academic researchers.
- The Ontario Ministry of Natural Resources and Forestry publishes requests for proposals (RFPs) on the provincial government public tender website. They have also placed a report on research needs, along with a brochure listing their

top 10 needs, on the Centre for Applied Science in Ontario Protected Areas website (<http://casiopa.mediamouse.ca/>).

Identifying Suitable Research Partners

For many COs, research partners were typically identified through their existing networks. For some organizations, this was the sole method of forging partnerships. Conservation organization staff members met researchers in person at workshops or conferences and have appointed researchers to their board of directors. Conservation organizations have also been approached by academic researchers seeking research opportunities and field sites. In some cases, it was challenging to find a match between the organization's research needs and the academic researcher's interests. No organization described a formal process for evaluating the quality of a potential research partner. Rather, this was done informally based on reputation within a network or the success of previous partnerships.

Efforts to identify a wider range of potential academic research partners would provide benefits similar to those derived from communicating research priorities. In this information age, the global research community is highly accessible and the need for in-person meetings, though highly beneficial, is reduced. Casting a wider net would allow COs to select the best partners based on expertise and prior successes. Furthermore, the process of approaching and evaluating a wide range of potential research partners would expose conservation practitioners to a variety of research that they might not otherwise discover and could lead to an expanded network of expert advisors. Conservation organizations could further refine their selection process by developing standards for evaluation. Standards could help in comparing candidates and even provide academic researchers with a template of skills to develop to improve their effectiveness in this role.

Examples of approaches for identifying suitable research partners:

- The Couchiching Conservancy in On-

tario has directly contacted global experts for specific research needs, thus targeting communication strategies to reach beyond their existing network.

- The Centre for Applied Science in Ontario Protected Areas at the University of Waterloo is a partnership between universities and COs that serves as a networking forum, facilitating knowledge transfer through workshops and connecting practitioners and researchers.

Requesting Research Products

Conservation organizations called for a wide range of products from academic partners, with formal reports frequently a standard requirement when funding or research authorization was provided. Some organizations believed that answers to specific management questions were the most important deliverable, whether they were provided orally or in written documents. Written documents included annual reports, student papers (theses, dissertations), literature reviews, technical reports, and peer-reviewed publications. Other important products included raw data, databases, maps, GIS shapefiles, models, and estimates of uncertainty, with the data often considered the most important deliverable. When direct funding was provided, expenditure reports were often required as well.

Conservation organizations valued each product differently, though some generalizations can be drawn. Peer reviewed papers were considered by many to be the most valuable deliverable because partners and stakeholders had more confidence in these products. When partnering with honors students, certain products had more value than others. For instance, literature reviews and the raw data generated through honors research were both described as valuable products, but management recommendations were often less suitable. The ability to interact with researchers was a key partnership benefit for many COs. Expert advice, presentations, and workshops were often highlighted as valuable contributions.

Ideally, deliverables should be of value

to both the CO and researcher. Reports may be useful for summarizing results and documenting completed research, but can be extra work that does not further researchers' career goals. Accepting peer reviewed publications as deliverables in lieu of reports can help alleviate this concern. Conversely, some publications that stem from research partnerships with COs are theoretical in nature and may be of limited value in informing management decisions. In such cases, a presentation or workshop in lieu of a report may be an efficient way for researchers to convey the type of information that practitioners need most. Workshops also provide a platform for questions, facilitating the presentation of complex information in a manner best suited for the particular audience.

Examples of research products valued by COs:

- The Land Management Branch of Alberta Tourism, Parks and Recreation primarily works with graduate students out of preference for peer-reviewed products; however, theses and prepublication reports are acceptable final products. They may require a 1–2 page report if results will not be published.
- The Nature Conservancy of Canada prefers peer reviewed publications with a practitioner as a coauthor, given that these improve the credibility of the organization and, therefore, facilitate fundraising.
- In addition to scientific products, the Couchiching Conservancy requests a plain-language summary of results to aid with outreach initiatives.
- The Manitoba Habitat Heritage Corporation has found that direct interaction with researchers helped to breach the ivory tower. Products that at first seemed irrelevant to management became valuable after researchers provided context and additional details.

Providing Research Support and Incentives

Direct funding is preferred by researchers but was often difficult for COs to provide. The amount provided ranged from a few hundred dollars derived from an endow-

ment fund to the full costs of a research project. Conservation organizations have covered the cost of field or travel expenses, researcher day rates, or partial or full graduate student stipends, among other research expenses. Leveraged or matching funds were often required by COs, as these can help COs to fundraise for research support. In the rare instances when large grants were available (typically from government organizations), a formal request for proposals (RFP) was usually developed and proposals were evaluated through a peer review process. ENGOs generally had more flexibility than government organizations in how they provided research support and sometimes provided unsolicited grants directly to researchers within their existing network.

Though COs (particularly ENGOs) could often only provide limited funding for research, they provided numerous other forms of support that are highly valued by researchers. Perhaps most importantly, many provided access to field sites (including park fee waivers). These sites are often unique in that they are relatively undisturbed and have no public access. Similarly, COs provided researchers with existing data sets or spatial layers, which also cost little to share. Access was provided to research stations, campsites, office space, and equipment; however, the availability of such resources depended on the needs of other practitioners or researchers. Staff, interns, and summer students occasionally contributed time towards research projects, including field support and technical expertise (e.g., GIS support, site-selection guidance). Though valued by researchers, CO staff member contributions were often limited by tight resources. Staff also contributed by facilitating permit requests, helping to develop research proposals to third-party funders, and guiding research through workshops, coauthorship, or as graduate student committee members.

Conservation organizations can entice researchers with a wide variety of incentives, many of which involve minimal cost to the organization. Access to field sites and data sets can be of immeasurable value to researchers and can be provided

with little effort. Occasional engagement and guidance from CO staff members can ensure that results are relevant (see “Ensuring relevance and quality of deliverables” section below). A less tangible, though no less important, contribution is the application by COs of research results in solving conservation problems—a benefit that carries personal appeal to many conservation researchers.

Examples of strategies for supporting research:

- The Nature Trust of British Columbia (TNTBC) developed the Brink/McLean Grassland Conservation Fund (\$2500/year) to support graduate student field expenses for research (not necessarily on their properties).
- Paying a day rate for academic partners to produce specific products (e.g., analyze data) was highly effective for TNTBC.
- Similarly, the Credit Valley Conservation Authority regularly offers contracts for researchers to provide workshops and summary reports on areas of expertise.
- Ducks Unlimited Canada contributes the services of GIS specialists and statisticians—along with an extensive library and librarian assistance—to their researcher partners.

Identifying Sources of Research Funding

Funding research by academic partners is an ongoing challenge for ENGOs. Governmental organizations generally had greater capacity to fund research through existing operating budgets and discretionary funds, but these sources are increasingly restricted. ENGOs typically had limited discretionary spending and avoided using it to fund partner-based research. Instead, in many cases, they fundraised directly for research. ENGOs sought to strategically match funders with specific projects of interest, including involving industrial partners that contributed to research initiatives. Several organizations developed comprehensive funding proposals for their work and included research costs as a component of the proposal.

Engaging in research partnerships may provide COs access to funding that they would not normally pursue, thus expanding their pool of donors rather than eating in to existing funding sources. Certain funding pools are only available to collaborative partnerships between researchers and practitioners. In other cases, the inclusion of practitioners as co-applicants, or at least a letter of support from a CO, can greatly enhance the chances of success for a grant proposal submitted by an academic partner. Conservation organizations with limited discretionary funds should consider such alternatives to fund research on critical knowledge gaps.

Examples of strategies for funding research:

- The Nature Conservancy of Canada strategically fundraises directly for research projects, while retaining scarce unrestricted funds for other purposes.
- Organizations such as Ducks Unlimited Canada and the Couchiching Conservancy include research as a component of funding requests. The development team of Ducks Unlimited Canada often includes research in requests for funding from corporations and private individuals.

Ensuring Relevance and Quality of Deliverables

Several organizations developed strategies to ensure that deliverables were received. Most commonly, deliverables were tied to funding or permitting requirements. When funding was provided directly, a certain percentage was frequently retained until final deliverables were received. If only permits were issued, similar recourse was not available but future permits could be withheld if previous deliverables were not received. For some organizations, specific partnerships were not renewed if deliverables were not received. For COs with limited capacity, it was often challenging to track receipt of required deliverables. Contacting partners via phone or email was usually an effective means for obtaining deliverables. Formal deliverables were not always a prerequisite of partnerships between COs and academic researchers.

Ensuring that deliverables are relevant to COs can be challenging, but several strategies are available to improve the relevancy of results. The best strategies included ongoing communication with researchers and direct involvement of practitioners in the research itself. The inclusion of practitioners on academic committees, as co-supervisors (via adjunct status), or as coauthors on technical reports and peer-reviewed publications, was highly effective. This fostered ongoing steering, ensuring that research remained relevant to COs.

Clearly describing required deliverables and timelines at project initiation helps to ensure completion. Some degree of flexibility and understanding is helpful when projects go awry or researchers face conflicting obligations. Developing systems and procedures to track reporting on partner-based research can help COs ensure that deliverables are received in a timely manner. Engagement in project scoping and design efforts can help to keep research on track. Hosting workshops and meetings throughout the data analysis and writing process can help to avoid topic drift, especially when field studies do not proceed as initially envisioned.

Examples of strategies for ensuring that deliverables are relevant and received:

- The Credit Valley Conservation Authority prefers, as a stipulation for graduate student funding, that students volunteer on a project for the organization in addition to their research duties (e.g., developing GIS layers of Emerald Ash Borer [*Agri-lus planipennis*] distributions that can be used by the organization for planning and outreach purposes).
- The Nature Conservancy of Canada (NCC) routinely asks researchers what NCC could do differently and considers this the most important deliverable. Students are particularly keen to engage in such opportunities to influence the operations of a major organization.

Ensuring that Research Results are Applied

Most research conducted in partnership between COs and academic researchers

effectively informed conservation actions. Some organizations believed that projects informed their work to a high degree, while for others this depended on the success and nature of individual research projects. Research results have been used to inform restoration efforts and guide securement priorities for land trusts. Human dimensions and social science research, such as surveys of park visitors, was increasingly viewed as informative to conservation practitioners. Several organizations reported that current/traditional management strategies were discontinued after research demonstrated their ineffectiveness.

Even basic research, not designed to address specific conservation issues, can provide benefits through helping practitioners better understand the systems they manage and by drawing attention to lands managed by the CO. Several COs included supporting research in their mandate, usually in the form of providing access to conservation lands. In such cases, results need not be immediately relevant to the organization's conservation practices. Generating basic ecological research is a sufficient result in and of itself.

Despite the numerous benefits of partner-based research, even applied projects do not always effectively inform conservation actions. Field research is unpredictable and weather events or equipment failure can lead to skewed results or insufficient sample sizes. Several COs noted that honors theses were frequently of limited value due to the limited experience of undergraduate-level students and short project times. For all academic research, land manager discomfort with advice from the ivory tower can limit knowledge transfer. A lack of time to review research can also limit the degree to which it is applied. Relevant research sometimes remains on the shelf due to lack of CO capacity.

Solutions exist to ensure that partner-based research is applied. Engagement by practitioners throughout the research project can ensure that knowledge is gained even when projects are unsuccessful. For honors studies, providing sufficient guidance and requesting tractable deliverables (e.g., survey results, other data) can ensure that

benefits are obtained. For all research, efforts by researchers to communicate results to practitioners can facilitate uptake. Workshops or meetings between researchers and managers throughout project implementation can help to break down language barriers, provide opportunities for questions, and ensure that results are used.

Examples of strategies for applying the results of partner-based research:

- The Land Management Branch of Alberta Tourism, Parks and Recreation considers research and management actions to be strongly linked, particularly because much of their partner-based research is applied and projects are often adjusted early on if problems occur. They translate research results into guidelines for best practices, convey research outcomes to upper management, and develop webinars and presentations from researchers to staff and other partners, including interpretive programs for visitor services programming staff.
- Environment Canada's Canadian Wildlife Service ensures that partner-based research is applied by citing it in updated management plans or recovery strategies. Internal practices (e.g., modeling exercises, designs of monitoring programs, and risk assessments) often involve the use of research results. Management actions are discontinued when research identifies strategies that are not effective.
- Ducks Unlimited Canada sees value in supporting, via fellowships, bright graduate students who are working on mission-related questions even if the research is more basic than applied. Outcomes still usually inform management. If they do not, it is often because Ducks Unlimited Canada staff members were not sufficiently engaged in guiding the research process.
- The Ontario Ministry of Natural Resources and Forestry policy staff members travel to internal offices to convey the results of visitor-use surveys to park managers. They also host monthly open calls with ecologists, providing a forum for sharing interesting results.

Measuring Success of Academic Partnerships

Though academic partnerships can be

highly beneficial to COs, few formal systems exist to measure the success of these partnerships, particularly at program or organization scales. Individual projects may be deemed successful if deliverables were received and results were used to inform conservation work; however, such evaluations are primarily informal. No examples were presented of formal frameworks for tracking the success of academic partnerships, though a few COs tracked the number of projects and resources involved.

Opportunities exist for COs to develop systems for tracking the success of academic partnerships. To review the success of academic partnerships, COs can begin by formally tracking partnerships and projects and the resources provided for and leveraged by these activities. Progress reports could detail the proportion and types of deliverables received and how these deliverables were used, including whether they were valuable in informing conservation activities. It may also be worthwhile to track the benefits to academic partners, such as which highly qualified personnel were trained and whether they later obtained conservation positions with the organization or other COs. Such systems can help COs revise how they manage partnerships, including troubleshooting areas where deliverables are not received or are not relevant and identifying best practices for managing partnerships and applying results. This information would be of value to development staff members seeking to justify research funding. Research products can be an important fundraising tool because they increase the credibility of the organization and its practices. Communication and outreach staff members seeking to celebrate the success of the organization in applying evidence-based conservation practices and the leadership of the organization in fueling conservation science could also apply data on successful research partnerships.

Examples of strategies for measuring the success of academic partnerships:

- The Ontario Ministry of Natural Resources and Forestry tracks the number of research authorizations issued each year

by discipline and evaluates whether the amount of research is increasing.

- Ducks Unlimited Canada supports graduate students, in part to ensure a steady supply of well-trained conservation professionals. They track the proportion of funded students that obtain conservation positions after projects are complete.
- Nature Canada conducts exit interviews with researchers to identify areas for improvement, lessons learned, and opportunities for future collaboration.

DISCUSSION

All COs interviewed engaged in academic partnerships and found great value in doing so. Each had developed unique approaches and strategies. Some generalities can be drawn regarding key components of a successful academic partnership program. A starting point is to develop and maintain a formalized list of research priorities and circulate this list beyond the CO's existing partner network. Conservation organizations should advertise the resources they can provide to academic research, including funding, access to lands and data, and other in-kind support. Formal requests for proposals followed by peer review of grant applications can help to ensure that high-quality partners and projects are selected, particularly when financial incentives are offered. Deliverables should be identified in writing and COs should follow up by phone or email to ensure receipt, holding back partial funding when possible until the project is complete. Peer reviewed publications are high-value deliverables to both COs and researchers, but other useful deliverables include reports, presentations, and workshop participation. Participation by CO staff members in the research process, through regular workshops or engagement as student committee members or coauthors, can help to ensure that the research focus remains relevant to the CO's information needs. Finally, COs should develop systems to track and apply partner-based research, as it both informs conservation work and supports communications and fundraising activities.

Though our review focused on Canadian organizations that manage lands for conservation, the results are broadly applicable to

other countries and types of COs. Further discussions with different types of COs, including those working at a global scale and with large academic and practitioner networks or in-house research networks, would yield additional opportunities for managing research partnerships. In the United States, for instance, The Nature Conservancy's science office is housed in the Biology Department of the University of Florida, facilitating partnerships among practitioners, researchers, and students. Similarly, the US Geological Survey (USGS) maintains long-standing Cooperative Fish and Wildlife Research Units (Coop Units), where they place researchers as faculty members at 40 academic institutions (<http://www.coopunits.org>). These government scientists have a reduced teaching workload, mentor graduate students, and can access federal grants and other resources. These partnerships allow for ample collaboration and resource leveraging between academia and the USGS, fueling considerable research on natural resources and their management. The Cooperative Ecosystem Studies Unit (CESUs) Network offers an even broader forum for partnership, encompassing 14 federal agencies in the United States and over 360 partners (<http://www.cesu.psu.edu>). Similar to the Coop Units, CESUs are placed at academic institutions, but include additional partners such as state agencies and NGOs. Their goal is to leverage partnerships to develop sustainability science that will inform decision-making. These collaborative efforts offer highly successful models for Canadian academic institutions and COs to consider.

Our review focused on the perspective of conservation practitioners; however, academic researchers have different considerations and perspectives on partnerships with COs. Cook et al. (2013) provide a summary of barriers and opportunities for academic researchers who wish to engage in applied conservation research. For academic researchers who are constrained by the publish-or-perish paradigm, some insight on how to focus partnership efforts can be gleaned from how COs value various research products. Peer-reviewed publications were highly valued, and

the delays and difficulties in publishing research were understood and accepted by COs. Conservation organizations were also often willing to accept publications or data sets in lieu of annual or final reports, easing the reporting burden on researchers. The high value of in-person expert advice suggests that researchers can contribute meaningfully to COs without major time commitments on activities that may fall beyond traditional academic reward structures; however, participation by researchers in conservation activities is among the most effective means of closing the research–implementation gap (Knight et al. 2008; Arlettaz et al. 2010).

Engaging in academic partnerships can lead to additional benefits for COs. Practitioners with advanced degrees can be awarded adjunct status, increasing the CO's profile and providing access to electronic journals and other university resources. Partners may elect to spend time with the CO as a visiting researcher or intern, thus contributing more directly to the CO's mission. Efficiencies can be gained through the development of multipartner programs where common research priorities are identified, CO resources are pooled, and researchers join forces to rapidly advance knowledge. Adopting a culture of collective learning, where practitioners, academic researchers, and other stakeholders collaborate to develop and implement applied research projects, may be the most effective means of generating knowledge that can improve conservation practices (Gonzalo-Turpin et al. 2008).

CONCLUSION

It can be challenging to find resources to develop and manage research partnerships, and to apply the knowledge gained from them. As with fundraising, a small investment in efforts to generate and manage research partnerships can lead to major returns. The solutions presented here do not require the development of a large research grant program; rather, they represent strategic options for allocating limited resources and funding in ways that are most effective and provide high rates of return. Efforts to document and communicate priorities

can expand the potential partner pool and help to uncover opportunities for resource sharing and collaboration, even among COs. Small grants helping to cover student stipends or researcher day rates can help facilitate active participation on applied issues for minor costs, and COs should remain nimble in offering alternative in-kind incentives. The review and application of partner-based research offers professional development opportunities and can improve staff knowledge, making this a worthwhile investment. Most importantly, engaging in an evidence-based model for conservation improves efficiency and effectiveness, ensuring better conservation outcomes and judicious spending of limited resources.

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Marie Tremblay has been with NCC since 2011, where she oversees NCC's conservation work in Alberta, including planning, securement, and stewardship activities on a land base covering over 200,000 acres. Marie possesses a Master's degree in Environmental Design and a PhD in ecology, both focusing on the movements of wildlife in fragmented landscapes. As a graduate student and a working professional, Marie possesses over 17 years of experience in conservation in Western Canada.

Heather Barna is the Director of Conservation Science and Planning for NCC. With a Master's of Science in community ecology and over 15 years of experience working with land trusts, she has a strong appreciation for conservation partnerships. As a botanist and conservation biologist, she has researched and implemented restoration projects throughout Canada and internationally.

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